

A prime-order obstruction to conjecture 00000008413

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Result. The claimed existence of Singer-parameter difference sets on noncyclic abelian groups for every prime-power parameter is false already at $n = 2$.

Original assertion and exact scope. Both original language versions assert that difference sets with parameters $(n^2 + n + 1, n + 1, 1)$ on noncyclic abelian groups exist if and only if n is a prime power. We refute the sufficiency direction. The original places no restriction excluding $n = 2$. We do not claim to settle the cyclic perfect-difference-set prime-power question or classify possible nonabelian realizations.

Standard difference-set definition. A $(v, k, 1)$ difference set in a finite group G is a subset $D \subseteq G$ satisfying

$$|G| = v, \quad |D| = k,$$

such that each nonidentity element $g \in G$ has exactly one ordered representation

$$g = ab^{-1}, \quad a, b \in D.$$

For $g \neq 1$, this automatically requires $a \neq b$. Thus the parameters concern the order of the ambient group, not merely the cardinality of the block or a subgroup. For Singer parameters they require

$$|G| = n^2 + n + 1, \quad |D| = n + 1.$$

Proposition 1. *No noncyclic group, hence no noncyclic abelian group, admits a difference set with Singer parameters at $n = 2$.*

Proof. The parameter $2 = 2^1$ is a prime power. Its Singer parameters are

$$(2^2 + 2 + 1, 2 + 1, 1) = (7, 3, 1).$$

Any alleged realization would therefore have ambient group order 7. Every group of prime order is cyclic: choose an element $g \neq 1$; its cyclic subgroup has order greater than one and dividing 7, so has order 7 and is the whole group. Thus the ambient group cannot be noncyclic, a contradiction. The difference-covering condition cannot overcome the absence of an eligible ambient group. $\square$

This is not failure of ordinary cyclic existence. The obstruction concerns the required noncyclic class. For comparison, in the additive cyclic group $\mathbb{Z}/7\mathbb{Z}$, the subset $\{0, 1, 3\}$ has the six ordered nonzero differences

$$0 - 1 = 6, \quad 0 - 3 = 4, \quad 1 - 0 = 1, \quad 1 - 3 = 5, \quad 3 - 0 = 3, \quad 3 - 1 = 2,$$

each occurring once. This elementary comparison is not needed for the disproof or its formal verification.

An independent issue with the claimed exception. The original also refers to a dihedral exception of group order 21. A finite dihedral group with m rotations has m rotations and m reflections, hence order $2m$. No such group has odd order 21. Calling the symmetry group of a 21-gon a dihedral group gives order 42, not 21. This is an additional necessary-condition obstruction, but the main disproof above does not depend on how that exception was named.

Actual objects and theorem scope in Lean. The project contains a faithful definition `SingerDifferenceSet` for actual finite subsets of actual groups. It records $\text{Nat.card } G = n^2 + n + 1$, the block cardinality $n + 1$, and unique ordered pairs for every nonidentity group element satisfying $ab^{-1} = g$.

- `PrimePower` uses the standard positive-exponent definition: $n = p^k$ for a prime p and $k > 0$. `two_prime_power` provides the witnesses $p = 2, k = 1$.
- `singer_two_forces_cyclic` derives actual ambient order 7 from the complete difference-set hypothesis and applies Mathlib’s prime-cardinality group theorem to obtain `IsCyclic G`.
- `no_noncyclic_realization` excludes every noncyclic ambient group, without even needing commutativity. Its abelian specialization covers the class specified in the original.
- `NoncyclicAbelianRealizable` existentially quantifies over ambient types, their actual commutative group structures, their noncyclicity, and the actual difference-set block.
- `counterexample` proves that 2 is a prime power but this existential realization statement is false. The conclusion is not conditional on a single preselected group.
- `no_dihedral_order_twenty_one` independently uses actual `DihedralGroup` cardinalities to exclude order 21.

No unique-difference property is assumed to be impossible on cyclic groups. It is the actual required ambient order together with noncyclicity that yields the contradiction. Proving nonexistence by a necessary defining condition suffices; enumerating hypothetical blocks in a nonexistent ambient group is unnecessary.

Reproduction. The project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. From `lean/`, run `lake build`, followed by:

```
lake env lean Main.lean -DwarningAsError=true.
```

Six principal theorem audits use only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no admitted proofs, additional axioms, native decision procedures, or external numerical computations.